

Problem 4

Problem 4

Problem 1 Let $f(x) = \sin(x)$ and $g(x) = \cos(x)$. Can you compute $f^{(48)}(x)$, $g^{(42)}(x)$, and $g^{(39)}(x)$?

Explanation. Let's find the first several derivatives of $f(x) = \sin(x)$.

$$\begin{aligned} f^{(0)}(x) &= \sin(x) \\ f^{(1)}(x) &= \cos(x) \\ f^{(2)}(x) &= -\sin(x) \\ f^{(3)}(x) &= -\cos(x) \\ f^{(4)}(x) &= \sin(x) \end{aligned}$$

Recall that, for n a nonnegative integer,

$$f^{(4n)}(x) = f(x) = \sin(x) \quad \text{and} \quad g^{(4n)}(x) = g(x) = \cos(x).$$

Thus:

$$\begin{aligned} f^{(48)}(x) &= f(x) = \sin(x) \\ g^{(42)}(x) &= g^{(40+2)}(x) = (g^{(40)})^{(2)}(x) = g^{(2)}(x) = g''(x) = -\cos(x) \\ g^{(39)}(x) &= g^{(36+3)}(x) = (g^{(36)})^{(3)}(x) = g^{(3)}(x) = g'''(x) = \sin(x) \end{aligned}$$